Luis Scheibe

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Education

California Polytechnic State University, San Luis Obispo, CA
Bachelor of Science in Mechanical Engineering, Expected Graduation December 2026
Master of Science in Mechanical Engineering, January 2027 - December 2027

Work Experience

Engineering Intern *August 2026 - Present*
Overwatch Aero, Santa Maria, CA

- Built and set up an Avid CNC machine, including all supporting infrastructure required for operation
- Designed and manufactured a mounting solution for a drone-mounted payload
- Designed and built a carbon fiber/composite cart with adjustable rollers
- Installed camera gimbals in helicopters and airplanes
- Designed a rooftop truck mount for a drone to validate pitot tube accuracy ahead of flight testing
- Obtained FAA Part 107 Remote Pilot Certification
- Built wiring harnesses for UAS platforms

Manufacturing Engineering Intern *June - August 2025*
Tri-Power Design, Denville, NJ

- Engineered and modeled an acetabular augmentation system prototype, applying DFM principles for revision total hip replacement
- Operated CNC machines, wire EDM, lathe, and manual mills to manufacture biomedical implants and equipment in compliance with ISO specifications
- Interpreted complex engineering drawings to successfully manufacture high-precision parts
- Assembled biomedical automation machines, ensuring functional performance against specifications

Engineering Project

Aerial Inspection Drone for Non-Destructive Testing *January 2026 - December 2026*
Voliro AG, San Luis Obispo, CA / Zurich, Switzerland - Industry-Sponsored Senior Design Project

- Developing a robotic arm payload to enable aerial EMAT area scanning on the Voliro T inspection drone
- Designed and validated complete payload assembly in SolidWorks, including structural architecture, mounting interface, and mass optimization
- Performed static and modal FEA to quantify stress, deflection, natural frequencies, and factor of safety under contact and flight loads
- Built and tested proof-of-concept prototype, integrating actuation, EMAT sensing, and control software for full system deployment

Leadership & Involvement

Tikkun Olam Makers (TOM), Cal Poly Chapter *Member*

- Design and 3D print affordable, open-source assistive technology for people with disabilities, including custom prosthetic attachments and mobility devices
- Collaborate with local partners including CAPSLO, SLO SELPA, and Cal Poly's Disability Resource Center to build devices for specific end users

Skills

Manufacturing & Prototyping: Engineering Drawings, Mill & Lathe Fabrication, 3D Printing (FDM/SLA), CNC Machining, Materials Selection & Testing
Programming: MATLAB, Python, Engineering Equation Solver (EES)
CAD/FEA: SolidWorks, ANSYS, Autodesk Fusion 360, Creo, Onshape, LTspice, Generative Design, Mass Reduction Analysis
Languages: English (fluent), German (fluent)